

The impact of logistics requirements and adjustments on firm performance: evidence from wood biomass supply chains

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ABSTRACT

Biomass wood plays a key role in renewable energy production. Using biomass for energy production reduces dependence on fossil fuels such as coal and oil, which are major sources of carbon dioxide emissions into the atmosphere. However, there are many challenges associated with biomass wood. One of them is logistics, especially storage and transportation requirements. Companies are investing in optimizing logistics processes to reduce operating costs and minimize environmental impact. The aim of this study is to develop and test the operationalization of logistics process requirements and the adaptation of logistics, and to empirically verify the impact of them on the performance of wood biomass companies. A survey of 300 wood biomass company respondents and structural equation modelling were conducted for the purpose of the article. Our research confirms a positive significant impact of logistics requirements enhanced by logistics adjustments on the firm's performance in wood biomass supply chain.

This study fills a gap in biomass logistics research by providing the first empirical model that tests how firms adapt their logistics to meet supply chain challenges. By focusing on firm performance, this work offers novel managerial insights into the strategic role of logistics adjustments within sustainable wood biomass supply chains.

Our results highlight the importance of adopting logistics practices to improve efficiency and provide valuable information to help managers and policymakers develop low-emission supply chain strategies.

1. Introduction

Biomass is a major source of energy, contributing to sustainable power generation, which plays a key role in tackling climate change and reducing greenhouse gas emissions. Millions of tons of biomass produced as a byproduct of commercial logging operations are used each year (Berry and Sessions, 2018; Hall, 2024). This is a very large market with its global value estimated at USD 127.4 billion in 2022 and is forecast to reach USD 203.2 billion by 2032, growing at a compound annual growth rate of 4.8 % (Takki, 2024).

One of its main advantages is that it is a very versatile energy source, generating not only electricity, but also heat and biofuels for use in many areas (Millinger et al., 2025). Increasingly, it is being used to extract valuable materials such as particleboard, fiberboard, composites or

products to serve the construction, furniture, food and pharmaceutical industries (Kawa and Pierański, 2023).

Biomass, like wood itself and related products, goes through numerous technological and logistics processes. In this case, we are discussing a reverse supply chain, where wood biomass is moved in the opposite direction of traditional raw material or product supply. This process is mainly related to storage and transportation (Kawa, 2023) which are significant cost factors (Malladi et al., 2018). For wood pellets, logistics costs can range from 34 % to 77 % of the total supply chain costs (WBA, 2018). In some cases, logistics costs account for up to 90 % of the total cost of raw materials (Malladi and Sowlati, 2018), underscoring the need for efficient supply chain management thanks to careful planning and efficient execution of processes are necessary (Kawa and Pierański, 2023).

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Companies are making investment efforts to optimize logistics processes and, at the same time, aim to reduce their environmental impact by reducing greenhouse gas emissions and waste. This includes the collection of residual wood biomass, as well as transportation, processing and reuse or recycling (Dubisz et al., 2023). These expenditures, while demanding, have the potential to reduce company costs while helping to reduce the amount of environmental waste generated by companies.

The management of residual wood biomass is an area of growing interest, and the number of empirical studies conducted in this area to date is relatively small. There are a wide variety of aspects and problems that can be studied in the context of reverse wood logistics processes. The management of wood biomass presents numerous challenges (Parajuli et al., 2023). In this context, there are issues related to different types of wood and their characteristics, processing technology, harvesting and transportation logistics. Analysis of different types of wood and their characteristics is important to adapt wood biomass processing to their specific characteristics. In addition, effective logistics of wood biomass requires consideration of its diversity and specific storage and transportation conditions (Searcy et al., 2014). Understanding and addressing these specific characteristics are key to effectively managing residual wood biomass and maximizing its potential in a sustainable manner, which is an important step toward more efficient and environmentally friendly waste management. Dynamic changes in environmental regulations and standards pose another challenge. Effective management of residual wood biomass, therefore, requires continuous adaptation to new regulations and standards to ensure compliance with environmental requirements (Page-Dumroese et al., 2022; Song and Aguilar, 2017).

Efficient management can significantly reduce operating costs and increase profits while also having a positive influence on the sustainable supply chain (Verma, 2024). Optimizing logistics processes allows for faster and more reliable delivery of items, which in turn improves customer satisfaction and loyalty. In addition, efficient inventory and transportation management minimizes losses and waste, and increases a company's flexibility and ability to respond quickly to market changes. As a result, well-organized logistics contributes to better performance and increased competitiveness (Grant et al., 2017).

To date, there have been no empirical studies that examine the relationship between wood biomass logistics and firm performance. In view of the findings presented, we posed the following research questions:

RQ1: How do logistics requirements and adjustments affect the performance of firms that handle wood biomass?

RQ2: How does logistics adjustments strengthen the impact of logistics requirements on the performance of firms that deal with wood biomass?

To answer these questions, we develop and test the operationalization of requirements for logistics processes and the adjustments of logistics to the requirements of biomass and empirically verify the impact of these requirements on the performance of wood biomass companies with mediation of logistics adjustments.

A quantitative survey was conducted as part of the study, preceded by focus group interviews and a literature analysis. A total of 300 respondents from companies that deal with wood biomass were surveyed.

Unlike prior studies that primarily focus on optimization and modeling of specific logistics components (Nunes and Silva, 2023; Parakhoo et al., 2022), our study introduces a novel empirical framework that captures the interaction between logistics requirements and adjustments. By linking these adjustments to firm performance, we contribute uniquely to the understanding of how sustainable logistics practices translate into tangible business performance.

2. Literature review

Biomass is of interest to many researchers. Recently, special

attention has been paid to research on wood biomass. It is dealt with by researchers associated with biology, chemistry, physics, but also computer science, operations research or logistics. The latter are important from the perspective of our article. Ferrarini et al. (2017) and Hrechyn et al. (2021) believe that although wood biomass logistics is an important topic, research in these areas is limited. The available studies lack a holistic view of logistics. It is usually considered pointwise, in one area.

In our work, we first focused on research in the field of wood biomass logistics in the previous 10 years (in the period from 2016 to 2025). A review of the Scopus and Web of Science databases based on the key term "wood biomass logistics" and focused on period between 2016 and 2025, resulted in 206 papers in the Scopus database and 335 papers in the Web of Science database. After filtering, we focused on 12 papers that directly investigate the area of logistics in the wood biomass industry. Their contribution in terms of geographical scope, research method(s), biomass type, variable for hypothesis testing and key findings are shown in Table 1.

While the logistics of wood biomass has attracted growing scholarly attention over the past decade, the prevailing body of research remains largely fragmented, heavily focused on technical optimization, and lacking in empirical grounding at the firm level. Most existing studies, such as those by Malladi and Sowlati (2018), Ko and Lautala (2018), and Han et al. (2018), emphasize the application of optimization models, simulation tools, or GIS-based planning to enhance specific logistics functions like transport routing, biomass densification, or multimodal infrastructure integration. Although these works have contributed valuable insights into logistics efficiency and environmental impact, they are often constrained to operational or engineering perspectives and rarely explore the managerial or organizational dimensions of logistics systems.

One of the most notable research gaps is the absence of empirically tested models that link wood biomass logistics to firm performance. Prior literature has typically stopped short of investigating how logistics challenges and adaptations translate into tangible outcomes for businesses operating within the biomass sector. Furthermore, many earlier works isolate and analyze discrete elements such as transport costs, storage losses, or the impact of moisture content, they seldom examine how firms holistically adapt their logistics networks in response to these constraints.

As many of the items in the literature refer to traditional biomass, it is worth highlighting the differences and similarities between them. The supply chains of wood biomass and traditional biomass are both central to the bioenergy sector but differ in structure and logistics (Guo et al., 2020; Junginger et al., 2011). Wood biomass is mainly sourced from forestry operations and follows a centralized, industrial supply chain with high infrastructure and moisture control requirements (Batidzirai et al., 2014; Parajuli et al., 2023). In contrast, traditional biomass, such as agricultural residues and organic waste, is collected in a decentralized manner and used mainly in local, small-scale energy systems (Junginger et al., 2011). Both supply chains face similar challenges, including seasonal availability, moisture sensitivity, bulk handling, and the need for regional sourcing and quality monitoring (Rentizelas et al., 2009; Sosa et al., 2015). Logistics optimization, such as efficient transport and storage solutions, is critical for maintaining biomass quality and reducing costs (Searcy et al., 2014).

In the next stage of our literature review, we used snowballing (Wohlin, 2014). We have identified several areas that relate to wood biomass logistics. They are described below.

2.1. Transport

A popular area in the literature is wood biomass transportation. This is most likely due to the fact that it is an expensive process. For this reason, it is common for researchers to analyze transportation costs, particularly operations and means of transportation. A study by Abbas et al. (2013) covered fixed costs for forest biomass removal operations,

Table 1

Overview of representative studies on wood biomass logistics (2016–2025), illustrating the diversity of research foci.

Study (year)	Region	Research Method/Survey	Biomass type	Variables for hypothesis testing	Key findings / contributions
Malladi & Sowlati (2018)	Global	Systematic literature review – optimization model classification and analysis	Wood biomass	N/A	Key logistics features (e.g. harvest, storage and transport) were identified in the models, as well as new trends. It was noted that, while recent models incorporate scattered feedstock availability, supply uncertainty, GIS integration and emissions, short-term planning and environmental/social objectives remain gaps.
Ferrarini et al. (2017)	Global	Three-stage systematic literature review, qualitative Impact Assessment (IA)	Mixed biomass including wood	N/A	The review focused on biomass logistics in the context of multipurpose bioenergy buffer strips. The importance of integrating ecosystem services with logistics planning was emphasised, with a particular focus on land-use planning as part of biomass supply strategies.
Berry & Sessions (2018)	Oregon, USA	Case study with modeling – Mixed-integer linear programming (MILP), facility costing model	Forest harvest residue	N/A	A mobile biochar facility was analysed against a centralised facility. It was found that periodically moving a small conversion plant (every 1–2 years) can reduce transport costs, with optimal mobility depending on the product (biochar or pellets) and the distribution of the feedstock. A trade-off was demonstrated between economies of scale and hauling distance.
Ko & Lautala (2018)	Michigan, USA	Optimization modeling – MILP	Wood residues (forrest, mill)	N/A	The evaluation involved the co-firing of coal power plants with woody biomass, with advanced logistics being employed, such as pelletizing and the use of custom rail cars. The best co-firing level was found, with the overall cost and emissions being minimised thanks to better feedstock logistics. It was shown that innovations in logistics (densification and multimodal transport) can increase the utilisation of viable biomass.
Han et al. (2018)	Pacific NW, USA	Optimization modeling – Mixed-integer programming	Forest residue	N/A	Forest residue collection was optimised on a tree-network road system. Recommended improvements to the road network and route planning, cutting transportation time and cost by ~ 10–15 %. The value of GIS-based tactical planning for biomass haulage was demonstrated.
Rodias et al. (2019)	Europe	Literature review – review classification by biomass type	Green, yellow, wood biomass	N/A	A comparison of supply chain management for green, yellow and wood biomass was made. The differences highlighted were that wood biomass chains often involve longer rotations and larger scales, and require the addressing of moisture and bulk issues. The literature was classified by the problems addressed versus the methodologies used, and it was pointed out that there is a need for more cross-cutting, advanced ICT (e.g. sensors, IoT) applications in biomass logistics.
Nunes et al. (2020)	Global	Literature review – review synthesis	Woody, green, yellow (general)	N/A	An up-to-date review of biomass supply chain modelling from the 2010 s was provided. It noted a maturation of models, but also fragmentation, with various approaches (e.g. MILP and simulation) and few standard benchmarks. It was recommended that future models should integrate uncertainty, policy and multi-criteria aspects to reflect real-world complexity.
Guo et al. (2020)	Spain	Business model analysis – MILP, cost analysis, scenario modeling	Forest biomass	N/A	Introduced the Integrated Biomass Logistics Center (IBLC) concept. Their model, applied to a Spanish feed mill, showed that using an IBLC to process biomass during idle periods can increase annual facility utilization and profit. First optimization framework for multi-use biomass supply chains, filling a conceptual gap and informing EU bioeconomy strategy.
Hrechyn et al. (2021)	Ukraine	Supply chain evaluation – observation, factor analysis and economic-mathematical modelling	Energy wood biomass	N/A	Developed a model evaluating economic and ecological performance of wooden biomass chains. Quantified an “eco-efficiency” indicator for biomass logistics. Found that optimizing routes and modernizing equipment improved both cost metrics and emissions per energy unit, illustrating achievable sustainability gains in Eastern European context.
Page-Dumroese et al. (2022)	USA	Narrative policy review	Forest biomass	N/A	The review examined U.S. policies for the utilisation of forest biomass. It was reported that wood biomass (e.g. logging residues) remains underused (with less than 5 % of residues in the south-eastern USA being used for pellets as of 2019) due to policy barriers and market factors. Policy gaps were identified, such as the need for infrastructure investment and consistent biomass definitions, as well as the need for streamlined regulations to improve logistics (e.g. weight limits for biomass trucks).

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Table 1 (continued)

Study (year)	Region	Research Method/Survey	Biomass type	Variables for hypothesis testing	Key findings / contributions
Ramos (2022)	Trans-Atlantic (EU–USA)	A qualitative, theory-driven case study	Industrial wood pellets	N/A	The wood pellet supply chain was examined as a “sociotechnical imaginary”, connecting power plants in the US Southeast and Europe. The discussion centred on the impact of narratives surrounding biomass on logistics investments and public acceptance, exploring the nuances of carbon-neutral fuel versus concerns over forest exploitation. While not quantitative, the study highlighted the interplay between social perception and logistics, for example, the pressure to certify sustainable sourcing adds complexity to supply chains.
Parajuli et al. (2023)	U.S. Southeast	Survey and industry data analysis – structured questionnaire and industry reports	Low-grade woody biomass	Harvesting practices, logistics issues, labor availability	The focus was on investigating the operational practices involved in harvesting wood biomass. Opportunities were found in adopting modern harvesting equipment and coordinated trucking, but challenges such as labour shortages and a lack of markets for low-grade biomass were also identified. It was emphasised that improving first-mile logistics (from forest to roadside) could significantly enhance supply chain efficiency in the region.
This paper	Poland	Quantitative research (survey) – structured questionnaire, structural equation modeling (SEM)	Wood biomass residues	Logistics requirements, logistics adjustments, firm performance	First study in biomass logistics literature to empirically model causal relationships between logistics requirements, logistics adjustments, and firm performance. Shifts the lens from “what the system should look like” to “how firms actually respond” to logistics demands. It provides quantitative managerial evidence that logistics decisions in biomass supply chains are strategic levers for performance.

Source: own elaboration based on Web of Science and Scopus databases.

pointing out the need for specialized equipment and considering the significant operational time spent on loading and unloading. According to the authors, research focusing on biomass transportation is limited, highlighting the need for further research in this area.

In the case of means of transport, the number of vehicles needed to fulfill a specific transportation need is considered. For example, a single type of truck or multiple types of trucks are analyzed (Malladi et al., 2018). A study by Shabani and Sowlati (2013) considered direct delivery of wood biomass from supply points to points without considering the option of storing biomass at intermediate facilities.

In terms of distances, types of transportation modes are analyzed. Wood biomass is transported long distances using railroads or barges due to their greater capacity compared to trucks. Of course, such transportation is contingent on access to railroads or waterways. Along with the transportation of biomass, Xie et al. (2014) also considered the transportation of biofuels from conversion plants to end customers using multiple modes of transportation.

Logistics in forestry has also been studied intensively for more than a decade in various regions. One of the main reasons for interest is to improve the economic viability of forest biomass utilization. In this regard, the best solutions are being sought for available resources, routing and distribution of wood biomass residues (Abbas et al., 2013).

Biomass logistics also looks at the impact of transportation on environmental issues, particularly CO₂ emissions (Ruiz et al., 2013). The main concern is the determination of optimal routes for transporting biomass and the use of appropriate means of transportation.

2.2. Costs

The use of wood biomass should be as efficient as possible (Sosa et al., 2015; Ko et al., 2018) to increase its use. The high cost of technology converting wood biomass to useful forms and supply chain costs are key impediments to its wider use (Malladi and Sowlati, 2018; Abbas et al., 2013; Frombo et al., 2009). Energy production from wood biomass is currently much more expensive than from traditional sources such as coal or gas. In addition, geographically dispersed biomass resources and the need to efficiently process them pose both financial and

environmental challenges (Han et al., 2018). The characteristics of biomass, such as its low density, make it less efficient to transport compared to fossil fuels (Allen et al., 1998). This also has implications for the environmental issues mentioned earlier (Ruiz et al., 2013).

Studies indicate that much of the cost of biomass energy production comes from logistics operations, and that is where cost savings should be sought. For example, a study in Denmark used GIS (geographic information system) to determine the cheapest strategies for allocating wood chips to power plants, minimizing transportation costs and time (Sosa et al., 2015). Other studies, such as those in Colorado, USA, have shown that optimizing biomass recovery operations can reduce logistics costs by up to 11 % compared to traditional systems (Han et al., 2018).

2.3. Methods and tools

Wood biomass logistics is of interest to researchers who are trying to improve it using tools and techniques. One of them is a mixed integer linear programming (MILP), which is used in a number of ways. It is relatively often used to solve transportation problems. An article by Zamora-Cristales et al. (2015) used it to perform simulations in quantifying the effects of truck-machine interaction, truck arrival rates and road conditions. Han et al. (2015) used MILP in combination with a network algorithm to solve a biomass logistics problem where the most favorable shredder locations and processing and transportation operations were unknown. The study showed that waste transportation is advantageous when biomass residues are relatively small or close to the processing site, while field shredding is more cost-effective for larger residue volumes. In contrast, Sosa et al. (2015) used MILP to analyze the impact of moisture content and truck configurations on supply chain costs and feedstock distribution.

Ko and Lautala (2018) used torrefaction for biomass pretreatment, reducing feedstock delivery costs and increasing co-firing. The MILP model was implemented to integrate the supply chain to find the optimal level of biomass co-firing with minimal transportation and logistics costs.

In turn, Ransikarbum and Pitakaso (2024) developed a MILP model to optimize the wood biomass supply chain by incorporating economic,

social, and environmental objectives. They integrated the model with the fuzzy analytic hierarchy process (FAHP) to address decision-makers' uncertainty and applied it to a real-world case in Thailand to analyze trade-offs and enhance strategic decision-making in biofuel logistics.

Simulation models are also used to analyze biomass. Sokhansanj et al. (2006) developed the Integrated Biomass Supply and Logistics (IBSAL) model to estimate the cost, at a given moisture content, of energy input and carbon emissions during the collection and transportation of agricultural biomass. The model has been adopted in several studies on agricultural biomass projects (Sokhansanj et al., 2009; 2010).

Another tool that is used in wood biomass logistics is the aforementioned GIS. It is used, among other things, to plot the geographic coordinates of the various elements of supply chains. This makes it possible, for example, to create a procurement plan that minimizes the cost of wood biomass (Keefe et al., 2014).

Frombo et al., 2009 proposed the Environmental Decision Support System (EDSS), which is GIS-based. It is used to better plan and manage the logistics of energy production from wood biomass, such as forest biomass, agricultural waste and industrial and municipal untreated wood residues.

Another technology that is increasingly used in industry and is partially replacing barcodes is RFID. It is used to identify things and read information without contact. So far, however, it has not been widely used in wood biomass logistics. Ranta et al. (2014) have conducted a demonstration study that uses RFID to manage energy biomass logistics.

In summary, biomass logistics is a complex and costly aspect of bioenergy production. Research focuses on improving and optimizing the entire supply chain, from harvesting and storage to transportation and processing, to make biomass a more viable energy source. Advances in technology, simulation models, and innovative logistics concepts are crucial for reducing costs and environmental impacts, thereby enhancing the sustainability and efficiency of biomass energy systems.

3. Hypotheses development

The appropriate use of logistics depends mainly on the specific characteristics of the production that results in biomass. These include weather conditions, environmental conditions, the activity of living organisms. One of the most important influencing factors is seasonality and the time factor, i.e. the need to perform work at the right time and during the right period, which in turn transfers to the availability of biomass. Seasonality implies unequal use of logistics resources, i.e., a large pile-up during the season, little and no use during the off-season (Tégla et al., 2012).

Biomass is so diverse that different parameters must be taken into account when collecting, preparing and transporting it. Due to its low energy density, it is necessary to transport it in much larger quantities compared to fossil fuels, leading to higher costs and higher CO₂ emissions resulting from transportation (Ruiz et al., 2013).

However, small elements such as sawdust can be converted into liquid fuels by thermochemical conversion, which makes transportation much easier (Stephen et al., 2010). Biomass is available in a dispersed manner in different geographic areas, which complicates collection and transportation operations. It is collected from different delivery points can have different quality characteristics, which affects its energy value (Malladi and Sowlati, 2018).

Depending on the type of biomass and climatic conditions of biomass storage, different types of storage facilities are used. The most common are indoor storage used for dry biomass and open storage used for biomass with higher moisture content. Long-term storage carries the risk of loss of quality and dry biomass, which can lead to additional costs and financial losses (Malladi and Sowlati, 2018).

The rate of biomass dry matter loss in storage is usually assumed to be constant in many mathematical models, but it actually increases at a decreasing rate. These losses increase the need for biomass in conversion

plants to meet requirements (Malladi and Sowlati, 2018).

Storing biomass for long periods of time can pose a fire risk due to the internal heat generated by the respiration of living cells in the biomass. The fire risk associated with long-term storage of biomass can also generate high insurance costs and safety measures (Malladi and Sowlati, 2018). Rentizelas et al. (2009) write about the risk of ignition of wood biomass residues during storage.

Keefe et al., 2014 consider that the quality and efficiency of biomass utilization is affected by many of its properties, such as the source of raw materials, their form, particle size, ash content or moisture content. The latter is particularly important. It affects how biomass is stored, transported and converted.

Moisture content is reduced through drying, which is an important quality criterion. The moisture content level affects the cost of drying, which is one of the key factors in optimizing costs in the wood supply chain (Godina et al., 2018). It is important that wood biomass for combustion has a uniformly low moisture content, which improves its energy value and combustion efficiency. Ash content also plays a key role. Wood biomass with a high ash content is considered less valuable because it leads to poorer combustion efficiency and generates additional costs for ash disposal (Greene et al., 2014). Appropriate harvesting methods make it possible to reduce both the moisture content and ash content of the delivered materials.

Biomass density and moisture content are important because of transportation regulations, which impose weight restrictions in many countries. Low calorific value, due in part to the high moisture content of most agricultural biomass, leads to higher harvesting, handling, transportation and storage costs. High or low moisture content may be desirable depending on the conversion process (Keefe et al., 2014).

One way to reduce the moisture content of biomass is torrefaction, which involves removing water and volatile organic compounds. It is used to reduce production costs and convert different types of biomass into a single product (Ko and Lautala 2018). Additional research is being conducted on torrefaction as a feedstock improvement technology to overcome biomass supply challenges. Svanberg et al. (2013) showed that torrefied biomass improves efficiency and reduces supply chain costs. Batidzirai et al. (2014), on the other hand, noted that while torrefaction currently results in higher biofuel costs, future improvements to the technology could significantly improve its efficiency.

Due to its physical properties, available raw biomass faces numerous obstacles in competition with fossil fuels (Godina et al., 2018). The transportation of wood in pellet form, while efficient, comes with numerous safety risks. Problems such as pellet decomposition during transport, the risk of dust and explosions, self-heating and ignition, and breathing difficulties pose significant challenges. Consequently, their unstable nature is an important factor in the biomass supply chain (Ramos, 2022).

In summary, biomass quality management, transport, storage and technological innovation are key to optimizing operating costs, increasing production efficiency and improving a company's competitiveness in the market. This can be associated with better firm performance (Malladi et al., 2018), which means higher profitability, profit, and sales revenue. Therefore, a research hypothesis has been put forward which is as follows:

- H1 Requirements for logistics processes in supply chain of wood biomass positively affect logistics adjustments.
- H2 Requirements for logistics processes in supply chain of wood biomass positively affect firm performance.

Numerous studies analyze different supply chain options (Anderson et al., 2012), taking into account the location of operations, biomass acquisition and processing, and transportation. Studies conclude that central processing is more economical due to economies of scale, reduced mobility and lower processing costs, although this is dependent on location and transportation costs (Berry and Sessions, 2018).

The physical properties of biomass, its location and distribution, as well as transportation infrastructure determine the scale and location of individual links in the supply chains. [Graham et al. \(2000\)](#) and [Elmore et al. \(2008\)](#) analyzed the impact of raw material availability and transportation costs on the scale of plants in a given region ([Stephen et al., 2010](#)).

The biomass supply chain differs from the large-scale crop supply chain. Collection takes place from small, often fragmented sites, and biomass transportation occurs along these patches of vegetation ([Ferrarini et al., 2017](#)). Setting up plantations close to the consuming plant is a fundamental way to reduce the cost and time of transporting raw material. However, an even more significant factor is the higher productivity of energy plantations compared to natural forest resources. Increased biomass production means less area is needed, which leads to shorter transport routes for fuelwood or raw materials ([Keefe et al., 2014](#)).

Depending on the source of the wood, the cost of acquiring biomass is often relatively low, sometimes even close to zero. However, the cost of collecting, processing, storing, and transporting it from the collection site to end-users can be quite high. In many cases, the logistics costs can exceed the value of the raw material itself. Therefore, optimizing the process of delivering wood to the processing plant is crucial for achieving the economic success of any biomass-related project ([Rentizelas et al., 2009](#)).

The cost of transportation depends on two main factors: distance and travel time. Distance mainly affects fuel consumption, while travel time determines costs related to depreciation, insurance, maintenance, and labor associated with the journey. Travel time includes both the round-trip travel time, especially when there is no backhaul load available, and the waiting time for loading and unloading ([Rentizelas et al., 2009](#)).

[Han et al. \(2018\)](#) point out that the benefits of transportation efficiency achieved through material densification via pre-processing can be negated if poor road conditions prevent large trucks from reaching the destination, significantly increasing transportation costs per unit. Therefore, careful selection of transportation means and the design of efficient raw material logistics are key factors for economically viable forest biomass supply. Designing efficient biomass supply logistics requires consideration of numerous parameters, such as the spatial distribution and quantity of biomass sources, road infrastructure and its condition, available equipment, and locations of storage and processing facilities.

Fleet management and the selection of vehicle types for transportation have been the subject of relatively many publications. These include strategies to increase system efficiency through fleet operations ([Ko et al., 2018](#)), such as truck scheduling based on travel time ([Ravula et al., 2008](#)), and outsourcing transportation ([Zhu et al., 2011](#)). Rail and air transport are usually provided as services by third parties (logistics companies) ([Hrechyn et al., 2021](#); [Rentizelas et al., 2009](#)), as they use very expensive transportation means unlike road transport. Various means of transport, such as trucks, trains, barges, or agricultural machinery, are used to transport biomass ([Stephen et al., 2010](#); [Ferrarini et al., 2017](#)). Trucks, which are commonly used for this purpose, prove to be cost-effective only over short distances. However, they are more flexible ([Rentizelas et al., 2009](#)) – they are more readily available and can move almost anywhere. Rail and barges are preferred for long routes and large volumes of biomass transported due to their cost-effectiveness. However, the use of these means of transport may be limited by the availability of supply and demand locations for transport by these specific means ([Malladi and Sowlati, 2018](#)).

It is suggested that savings can be achieved by using larger trucks for long distances and smaller trucks for shorter routes. Additionally, direct delivery of biomass from suppliers to customers, bypassing warehouses, can lead to cost reductions ([Malladi et al., 2018](#)).

Plantations and forests on flat terrain, close to end-users, with access to paved roads with high load-bearing capacity are a good source for minimizing transportation costs. However, forest areas often have only

natural, steep, narrow, and winding roads of low standard, which makes it difficult for long, low-slung tractor-trailers to access ([Keefe et al., 2014](#)).

It is also worth noting the limitations of transport means. Maritime transport is limited by volume for all raw materials except bio-oil, which is weight-limited. In the form of bio-oil, three times more energy can be delivered than wood chips using the same number of ships. However, in truck transport, which is weight-limited, the benefits of densified intermediate raw materials, such as bio-oil and pellets, are much smaller ([Stephen et al., 2010](#)). Therefore, changing the state of biomass or its densification is important for the efficient use of the transport fleet ([Han et al., 2018](#)). Often, there are constraints requiring the use of specific types of trucks for transporting different types of biomass due to differences in their size and properties. Not all types of trucks can reach all locations in the supply chain due to restrictions on vehicle size and available space in specific locations ([Malladi et al., 2018](#)).

The above actions and processes require investment in advanced systems and equipment, which may increase costs in the short term but improve efficiency and reduce losses in the long term. This allows for cost optimization of processing and logistics, which can lead to increased profitability of operations and better resource utilization. This has an impact on the company's results.

The aforementioned observations are the basis for another research hypotheses:

- H3 Logistics adjustments in supply chain of wood biomass positively affects firm performance.
- H4 The impact of requirements for logistics processes in supply chain of wood biomass on the firm's performance is strengthened by logistics adjustments.

The assumption about the existence of specific relationships expressed in the form of the hypotheses is the basis for the model which presumes co-existence of the constructs and their joint influence on firm performance. A graphic form of the model has been presented in [Fig. 1](#).

4. Methodology

4.1. Research process

The paper presents the results of quantitative research, preceded by a literature review and Focus Group Interviews (FGI) ([Kawa 2023](#); [Kawa and Pierański, 2023](#)). The interviews facilitated an understanding of issues related to the reverse supply chain of wood biomass and consequently helped in preparing the survey questionnaire.

We used a Computer-Assisted Telephone Interviewing (CATI) survey as the data collection tool. The research was conducted by an external research agency specializing in both qualitative and quantitative studies. Trained interviewers carried out the interviews between November and December 2022, using a scenario and questionnaire

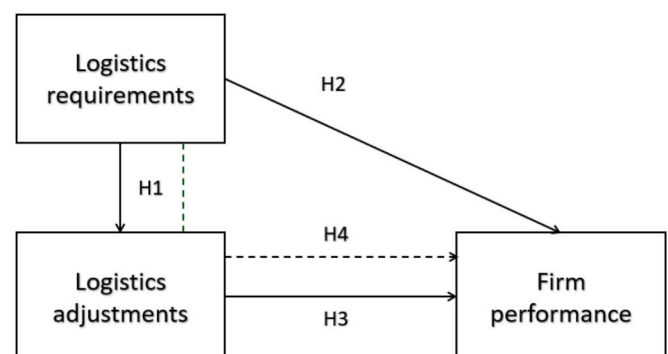


Fig. 1. Model of wood biomass logistics and research hypotheses.

developed by the research team. We recognise that some time has elapsed since the study was conducted, but unlike sectors subject to rapid technological disruption, wood biomass logistics is characterised by slower structural change. Operational constraints and challenges tend to evolve incrementally. As a result, the 2022 data remain relevant and reflects current industry dynamics.

Participants were selected using stratified random sampling, focusing on companies dealing with wood biomass residues. Given the nature of the research, the general population was narrowed down to enterprises involved in wood biomass residues, including: production or processing of wood or wood-based products; recycling, disposal, and processing of wood waste; transport or storage of wood waste.

The database included approximately 12,000 entities. The research agency contacted about 3,800 individuals. Of these, approximately 3,500 refused to participate, resulting in 300 participants and a success rate of 8 %. This sample is representative of a population of up to 50,000 entities in Poland, with a margin of error of 5.64 % (Raosoft, 2025).

To participate in the study, respondents needed experience in the wood industry, particularly with biomass. This was ensured through two screening questions: whether their company deals with wood residues, wood materials, or post-consumer wood, and whether they have knowledge about the generation, processing, collection, or disposal of wood waste. Only those who answered affirmatively to both questions were included in the study.

The questionnaire consisted of three parts: an introduction, main questions, and a demographic section. The introduction provided detailed information about the project title, research purpose, interview procedures, and confidentiality assurances. The demographic section helped characterize the respondents and their companies.

4.2. Operationalization of the variables

The wood biomass logistics model shown in Fig. 1 consists of three constructs, which are latent variables. Due to the deficit of empirical studies, signaled previously, both in the field of logistics wood biomass, the indicators of these variables have to be prepared. Observable indicators, which have been included in the questionnaire in the form of statements, were developed for each of the latent variables (Table 2).

With the use of the exploratory factor analysis the indicators with the highest loading values have been found. This has caused, on the one hand, reduction of the indicators and, on the other hand, better statistical adjustments of the factors.

The construct “logistics requirements” consists of six indicators. They were developed on the basis of studies by such authors as Ko and Lautala (2018); Greene et al. (2014), Téglá et al. (2012); Keefe et al. (2014). Seven indicators make up the construct “logistics adjustments” (Table 1). The content of the indicators was developed from the studies by Ruiz et al. (2013); Guo et al. (2020), Rentizelas et al. (2009). The rating scale used for this construct has obtained a positive result in the reliability test. The content of the questionnaire statements, the relevant codes assigned to them, the results of the reliability tests (Cronbach’s α , AVE and CR) and the scale used are presented in Table 1. The statements have been evaluated by the respondents with the use of a five-point Likert scale, where 1 was “not important at all” and 5 was “very important”.

The last construct included in the model is “firm performance”, which consists of: market share, sales revenues, profitability, profit, new products/services (Table 1) (Fynes and Voss 2002; Homburg et al., 2004). The respondents were asked to compare their performance with those of their direct competitors in the last financial year, using a scale of 1–5, where 1 meant “much worse” and 5 meant “much better”. Although such a comparative approach to the evaluation of the indicators may raise questions about the validity of use, it is increasingly being applied in research (Fynes and Voss 2002; Homburg et al., 2004).

Table 2

The measurement of wood biomass logistics variables.

Code	Statement	α	AVE	CR
Logistics requirements (Ko and Lautala, 2018; Greene et al., 2014; Téglá et al., 2012; Keefe et al., 2014)				
q19s1	Sources and sourcing of wood biomass residues	0,780	0.407	0.804
q19s2	Origin, structure, and composition of wood biomass residue			
q19s3	Required moisture content of the wood biomass residue and/or product			
q19s4	Required temperature of the wood biomass residue and/or product			
q19s5	Physical form of the wood biomass residue (dimensions, volume, weight, etc.)			
q19s6	Degree of risk of ignition of the wood biomass residue			
Logistics adjustments (Ruiz et al., 2013; Guo et al., 2020; Rentizelas et al., 2009)				
q21s1	Configuration of the logistics chain for organizing the flow of biomass residues	0.866	0.478	0.864
q21s2	Method of storage for wood biomass residues			
q21s3	Timing of transportation of residual wood biomass			
q21s4	Timing of transportation of processed product			
q21s5	Selection of logistics units dedicated to transportation and/or storage			
q21s6	Selection of types of vehicles to carry out transports			
q21s7	Selection of additional safeguards			
Firm performance (Fynes and Voss 2002; Homburg et al., 2004)				
q30s1	Market share	0.855	0.550	0.855
q30s2	Sales revenue			
q30s3	Profitability			
q30s4	Profit			
q30s5	New products/services			

5. Results

The next research stage has been to construct the measurement model and, then, the structural one. The AMOS software has been used for modeling. To test the importance of the measurement model and assess its quality, the confirmatory factor analysis, which consists in combining the given indicators with the constructs measuring them, has been applied.

The last stage of the modeling has been to prepare the structural model (Fig. 2), in which the causal relations among the latent constructs are tested. To assess the model, validity measurements were used which are necessary to estimate structural equations models. All the values obtained are at a satisfactory level. The standardized χ^2 (χ^2/ss) is 1.622 and is much lower than the maximum value of 5. The root mean square error of approximation (RMSEA) is 0.046, which proves good adjustments of the model. LO 90 is close to 0 (0.034), HI 90 is less than the maximum of 0.08 (0.057). The goodness of fit index (GFI) is 0.932, and the adjusted goodness of fit index (AGFI) is 0.903, which is above the limit of a well matched model (0.9). The values of the incremental indexes also indicate a very good match of the model to the data. The normed fit index (NFI), the incremental fit index (IFI) as well as the Tucker-Lewis index (TLI) and the comparative fit index (CFI) are well above the minimum value of 0.9. All of them are above 0.96 (see Table 3). Very good fit indices confirmed the suitability of the model to the data.

The analyzed model has obtained satisfactory measures of adjustments to the data from the empirical studies. This has been the basis for further actions, especially for the evaluation of the research hypotheses. Table 4 shows the parameters for the paths specified in our model. There have been no bases to reject hypothesis H1, which presumed a positive

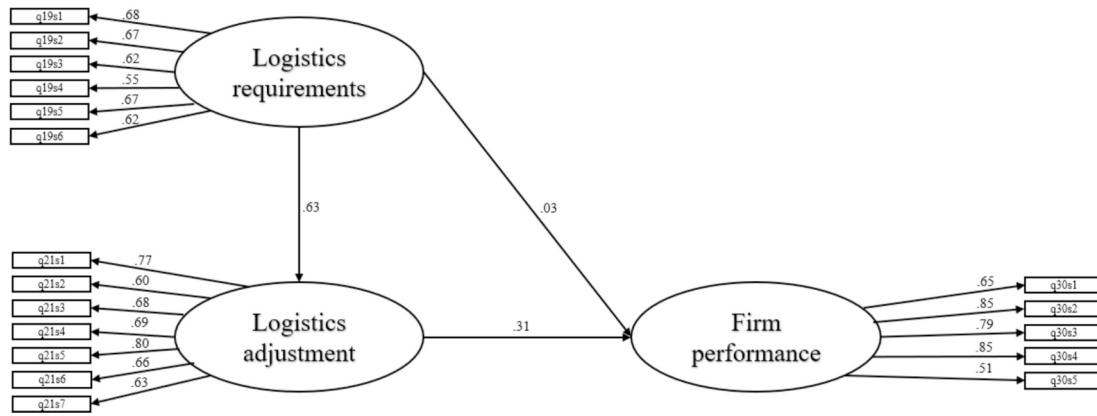


Fig. 2. Structural model of wood biomass logistics.

Table 3

Fit indices of the wood biomass logistics structural model.

χ^2/df	GFI	AGFI	NFI	IFI	TLI	PGFI	PNFI	RMSEA	LO 90	HI 90	CFI
Tolerance range <2 or <5 p < 0.01	≥0.9	≥0.9	≥0.9	≥0.9	≥0.9	>0.6	>0.6	<0.05 p < 0.05	Close to 0	<0.08	≥0.9
Structural model of wood biomass logistics 1.622 p < 0.001	0.932	0.903	0.921	0.968	0.959	0.654	0.723	0.046	0.034	0.057	0.968

Table 4

Relationship between constructs in the structural model.

Path (hypothesis)	Standardized parameter
Logistics requirements → Logistics adjustments (H1)	0.628***
Logistics requirements → Performance (H2)	0.026
Logistics adjustments → Performance (H3)	0.307**
Logistics requirements → Logistics adjustments → Performance (H4)	—

p < 0.01; *p < 0.001.

influence of logistics requirements on logistics adjustments. This dependence has been statistically significant and strong. Contrary to expectations, hypothesis H2 has not been supported. Although logistics requirements have a positive influence (very small) on firm performance, this correlation has not been statistically significant. The results of the studies have also demonstrated a statistically significant effect of logistics adjustments on performance (hypothesis H3). The last hypothesis (H4) that the impact of logistics requirements on firm performance is reinforced by logistics adjustments has been positively tested.

Although all of the model fit indices obtained are at a satisfactory level, this does not mean that the model developed fully reflects the economic reality and guarantees the truth of the theory. The abductive approach used in the paper allows to indicate the most probable hypotheses in a selected fragment of the reality.

6. Discussion

Studying the logistics of wood biomass is extremely important from the perspective of sustainable development and operational efficiency of companies that deal with wood biomass. Our study aimed to determine the impact of logistics requirements on firm performance, as well as the role of logistics adjustments as a mediator in this relationship.

Wood biomass logistics involves a number of challenges, such as the diversity of wood raw materials, processing, collection and transport

requirements, as well as dynamically changing environmental protection regulations. Effective management of this category of waste requires understanding the specific characteristics of biomass, which allows for better adaptation of logistics processes.

The process of structural modeling allows to refer to the hypotheses assumed in this paper at the final stage. Table 5 presents a summary of the effects of the research hypotheses testing. Three of the four hypotheses were supported: hypotheses 1,3 and 4. Hypothesis 2, despite the relationship in line with the expected direction, did not achieve statistical significance. The tests of the hypotheses were based on estimated parameters of the finally approved structural model.

Hypothesis H1 assumed that the requirements regarding logistics processes have a positive impact on the adjustments of the company's logistics. The research results confirmed this hypothesis, indicating a statistically significant and strong relationship between the two variables. According to the literature mentioned above, various factors in

Table 5

Evaluation of the research hypotheses.

Hypotheses	Result of hypothesis testing	Comments to hypothesis
H1. Requirements for logistics processes in supply chain of wood biomass positively affect logistics adjustments	Supported	Statistically significant positive influence
H2. Requirements for logistics processes in supply chain of wood biomass positively affect firm performance	Rejected	Statistically insignificant positive influence
H3. Logistics adjustments in supply chain of wood biomass positively affects firm performance	Supported	Statistically significant positive influence
H4. The impact of requirements for logistics processes in supply chain of wood biomass on the firm's performance is strengthened by logistics adjustments	Supported	Statistically significant positive influence

the organization and course of logistics processes carried out in relation to wood biomass residues influence the realization of logistics processes, such as transport efficiency, storage management, processing or the configuration of the supply chain for organizing the flow of wood biomass residues.

Hypothesis H2, which assumed a direct impact of logistics requirements on firm performance, was not supported by the research results. Although logistics requirements showed some positive impact, they were not statistically significant. This may suggest that logistic requirements alone, without appropriate logistics adjustments, are not sufficient to achieve better results. This is consistent with previous research that emphasizes that the logistics system itself must be dynamically adapted to changing conditions and market requirements to bring tangible benefits (Pfohl & Buse, 2000). Even the best-managed logistics requirements may turn out to be ineffective if they are not supported by appropriate adaptation of logistics processes. It should be noted that logistics adjustments may not be the only mediating factor. Other variables, such as technological capabilities, regulatory compliance strategies, human resource competencies, or firm-specific characteristics such as size and market agility may also play moderating or mediating roles in translating logistics demands into firm performance. Future research could extend our model by incorporating these contextual or organisational contingencies to better understand performance drivers in wood biomass logistics.

Hypothesis H3 concerned the impact of logistics adjustments on firm performance. Research has shown that logistics adjustments has a significant and positive impact on the performance of wood biomass companies. The literature emphasizes that flexible and well-managed logistics processes can lead to improved firm performance thanks to better coordination of activities, cost reduction and increased operational efficiency (Tracey et al., 2005). In particular, optimization of biomass logistics, including collection, transport and processing, is a key factor influencing the costs and efficiency of the entire supply chain.

Hypothesis H4 assumed that logistics adaptation enhances the positive impact of logistics requirements on firm performance. The research results confirmed this hypothesis, indicating that logistics adjustments plays a mediator role in this relationship. There are many examples in the literature confirming that the integration and adaptation of logistics processes to specific operational requirements can lead to improved company results (Prajogo et al., 2016), Romano (2003). For example, optimizing the logistics network for biomass, taking into account specific equipment and logistics requirements, leads to cost reductions and increased operational efficiency. Companies that effectively integrate their logistics processes with market requirements can better respond to changing conditions, which translates into better performance.

Our contribution is novel in that it introduces a validated empirical model based on a large-scale survey that reveals the mediating role of logistics adjustments. This concept is rarely operationalised in biomass logistics literature. This provides managers with actionable insights into where to focus their adaptation efforts to improve performance, moving beyond traditional cost- or technical-based optimisation frameworks.

In response to our initial research question (RQ1), we find that logistics requirements alone do not significantly improve firm performance; rather, their effectiveness depends on the firm's ability to adapt logistics processes. Logistics adjustments independently demonstrates a robust positive effect on performance. It acts as a mediator, significantly amplifying the influence of logistics requirements on firm performance (RQ2). This mediating role highlights that effective adaptation and integration of logistics processes into strategic and operational frameworks are crucial for translating logistics requirements into performance improvements.

These findings suggest that, in addition to implementing storage and handling standards, for example, managers should focus on continuously adapting logistics processes to meet these requirements and changing conditions. It is this adaptability and integration that yield performance gains.

7. Conclusions

Wood biomass logistics is a complex and multi-faceted process that requires a coordinated approach. Our research shows that logistics requirements enhanced by logistics adjustments have a significant impact on the performance of wood biomass companies. Effective management of biomass transport and storage, investing in advanced logistics technologies and process optimization are key to improving the operational and financial results of companies.

This study is novel in that it shifts the analytical focus from idealised logistics configurations to firm-level behavioural adaptation. This provides a fresh approach to sustainable logistics, showing that internal adjustments, rather than external constraints alone, influence business performance in biomass supply chains.

Our research provides contributions for both theoreticians and practitioners, pointing to the need for further research and investment in effective wood biomass logistics.

7.1. Theoretical and practical implications

The research results largely confirm the findings of the literature, pointing to the key role of logistics and its adaptation in improving firm performance. Empirical research shows that logistics has a significant impact on the operational and financial performance of wood biomass companies. These findings are consistent with previous research that highlights that effective logistics management, especially in the biomass context, requires flexibility, technological innovation, and process integration to achieve better business results.

Our research makes an important contribution to the literature in the field of wood biomass logistics by showing that logistics adjustments is a key factor mediating the impact of logistics requirements on firm performance. These results support the idea that effective logistics management in the biomass context can lead to significant operational and financial benefits.

From a practical perspective, our research provides valuable guidance for managers of wood biomass companies. Firstly, it is crucial that companies invest in advanced logistics systems and technologies that will enable effective adaptation of logistics processes to the specific requirements of biomass. Second, transportation optimization, including the selection of appropriate transportation modes and biomass density, can significantly reduce operating costs and increase efficiency. Finally, warehousing management, including quality control and warehouse space efficiency, is crucial to minimizing losses and improving operational performance.

Moreover, policymakers should consider supporting infrastructure and incentives for low-emission biomass transportation, as well as clear guidelines that encourage firms to invest in adaptive logistics practices, given their proven impact on performance and sustainability.

In summary, the main contributions of the research are as follows:

- Empirical validation of the mediating role of logistics adjustments in supply chain of wood biomass.
- Demonstration of the limited direct impact of logistics requirements without appropriate adjustments.
- Identification of key logistics factors influencing firm performance in the context of wood biomass supply chains.
- Extension of the logistics performance literature specifically tailored to the unique context of wood biomass supply chain.
- Providing actionable insights for managers in wood biomass logistics for enhanced strategic and operational decision-making.

7.2. Research limitations and future research directions

Despite the significant results, our study has some limitations. Firstly, the research was conducted on a sample of 300 companies – although it is sufficient to draw conclusions, it may not reflect the full

diversity of companies dealing with wood biomass. Second, the research focused on companies operating in one country, which may limit the generalization of the results to other geographic and economic contexts. These limitations can be addressed with further research. Furthermore, our research highlights the need for studies on specific aspects of biomass logistics, such as storage and transportation management, to better understand their impact on company performance.

Additionally, future research should seek to investigate the impact of other factors, such as legal regulations and government policies on wood biomass logistics. In addition, it is worth researching new technologies and innovations in biomass logistics, which may contribute to further optimization of processes and improvement of company results.

CRediT authorship contribution statement

Arkadiusz Kawa: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Davor Dujak:** Writing – review & editing, Resources, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Arkadiusz Kawa reports financial support was provided by National Science Centre Poland. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- Abbas, D., Handler, R., Dykstra, D., Hartsough, B., Lautala, P., 2013. Cost analysis of forest biomass supply chain logistics. *J. For.* 111 (4), 271–281. <https://doi.org/10.5849/jof.12.054>.
- Allen, J., Browne, M., Hunter, A., Boyd, J., Palmer, H., 1998. Logistics management and costs of biomass fuel supply. *Int. J. Phys. Distrib. Logist. Manag.* 28 (6), 463–477. <https://doi.org/10.1108/09600039810245120>.
- Anderson, N., Chung, W., Loeffler, D., Jones, J.G., 2012. A productivity and cost comparison of two systems for producing biomass fuel from roadside forest treatment residues. *For. Prod. J.* 62 (3), 222–233. <https://doi.org/10.13073/0015-7473-62.3.222>.
- Batidzirai, B., van der Hilst, F., Meerman, H., Junginger, M.H., Faaij, A.P., 2014. Optimization potential of biomass supply chains with torrefaction technology. *Biofuels Bioprod. Biorefin.* 8 (2), 253–282. <https://doi.org/10.1002/bbb.1458>.
- Berry, M.D., Sessions, J., 2018. The economics of biomass logistics and conversion facility mobility: an Oregon case study. *Appl. Eng. Agric.* 34 (1), 57–72. <https://doi.org/10.13031/aea.12383>.
- Dubisz, D.K., Augustyniak-Wysocka, D., Koliński, A., 2023. Efficiency of reverse wood biomass supply chains – a qualitative approach. *Zeszyty Naukowe Politechniki Poznańskiej*. <https://doi.org/10.21008/j.0239-9415.2024.090.08>.
- Elmore, A.J., Shi, X., Gorence, N.J., Li, X., Jin, H., Wang, F., Zhang, X., 2008. Spatial distribution of agricultural residue from rice for potential biofuel production in China. *Biomass Bioenergy* 32 (1), 22–27. <https://doi.org/10.1016/j.biombioe.2007.06.005>.
- Ferrarini, A., Serra, P., Almagro, M., Trevisan, M., Amaducci, S., 2017. Multiple ecosystem services provision and biomass logistics management in bioenergy buffers: a state-of-the-art review. *Renew. Sustain. Energy Rev.* 73, 277–290. <https://doi.org/10.1016/j.rser.2017.01.052>.
- Frombo, F., Minciardi, R., Robba, M., Rosso, F., Sacile, R., 2009. Planning woody biomass logistics for energy production: a strategic decision model. *Biomass Bioenergy* 33 (3), 372–383. <https://doi.org/10.1016/j.biombioe.2008.09.008>.
- Fynes, B., Voss, C., 2002. The moderating effect of buyer-supplier relationships on quality practices and performance. *Int. J. Oper. Prod. Manag.* 22 (6), 589–613. <https://doi.org/10.1108/01443570210427640>.
- Godina, R., Nunes, L.J., Santos, F.M., Matias, J.C., 2018. Logistics cost analysis between wood pellets and torrefied biomass pellets: The case of Portugal. In 2018 7th International Conference on Industrial Technology and Management (ICITM) (pp. 284–287). IEEE. doi: 10.1109/ICITM.2018.8333962.
- Graham, R.L., English, B.C., Noon, C.E., 2000. A geographic information system-based modeling system for evaluating the cost of delivered energy crop feedstock. *Biomass Bioenergy* 18 (4), 309–329. [https://doi.org/10.1016/S0961-9534\(99\)00098-7](https://doi.org/10.1016/S0961-9534(99)00098-7).
- Grant, D.B., Wong, C.Y., Trautrim, A., 2017. *Sustainable Logistics and Supply Chain Management: Principles and Practices for Sustainable Operations and Management*. Kogan Page Publishers.
- Greene, W.D., Cutshall, J.B., Dukes, C.C., Baker, S.A., 2014. Improving woody biomass feedstock logistics by reducing ash and moisture content. *Bioenergy Res.* 7, 816–823. <https://doi.org/10.1007/s12155-013-9404-6>.
- Guo, X., Voogt, J., Annevelink, B., Snels, J., Kanellopoulos, A., 2020. Optimizing resource utilization in biomass supply chains by creating integrated biomass logistics centers. *Energies* 13 (22), 6153. <https://doi.org/10.3390/en13226153>.
- Hall, P., Residual biomass fuel projections for New Zealand: 2024 – Indicative availability by region and source. <https://www.ieabioenergy.com/wp-content/uploads/2024/11/NZ-Woody-Biomass-Residues-and-Resources-2024.pdf>.
- Han, H., Chung, W., Wells, L., Anderson, N., 2018. Optimizing biomass feedstock logistics for forest residue processing and transportation on a tree-shaped road network. *Forests* 9 (3), 121. <https://doi.org/10.3390/f9030121>.
- Homburg, C., Krohmer, H., Workman Jr., J.P., 2004. A strategy implementation perspective of market orientation. *J. Bus. Res.* 57, 1331–1340. [https://doi.org/10.1016/S0148-2963\(03\)00069-9](https://doi.org/10.1016/S0148-2963(03)00069-9).
- Hrechyn, B., Krykavskyy, Y., Binda, J., 2021. The development of a model of economic and ecological evaluation of wooden biomass supply chains. *Energies* 14 (24), 8574. <https://doi.org/10.3390/en14248574>.
- Junginger, M., van Dam, J., Zarrilli, S., Mohamed, F.A., Marchal, D., Faaij, A. (2011). Barriers and opportunities for global bioenergy trade. In *International bioenergy trade: History, status & outlook* (pp. 69–94). Springer. doi: 10.1007/978-94-007-0705-4_2.
- Kawa, A., 2023. Reverse supply chain of residual wood biomass. *LogForum*, 19(2), [Page numbers if available]. doi: 10.17270/JLOG.2023.799.
- Kawa, A., Pierański, B., 2023. Wood biomass in supply chain – evidence from the FGI studies. *Bus. Logist. Modern Manage.* 23, 373–389.
- Keefe, R., Anderson, N., Hogland, J., Muhlenfeld, K., 2014. Woody biomass logistics. In *Cellulosic energy cropping systems* (pp. 251–279). John Wiley & Sons, Ltd. doi: 10.1002/9781118676332.ch14.
- Ko, S., Lautala, P., 2018. Optimal level of woody biomass co-firing with coal power plant considering advanced feedstock logistics system. *Agriculture* 8 (6), 74. <https://doi.org/10.3390/agriculture8060074>.
- Malladi, K.T., Sowlati, T., 2018. Biomass logistics: a review of important features, optimization modeling and the new trends. *Renew. Sustain. Energy Rev.* 94, 587–599. <https://doi.org/10.1016/j.rser.2018.06.052>.
- Malladi, K.T., Quirion-Blais, O., Sowlati, T., 2018. Development of a decision support tool for optimizing the short-term logistics of forest-based biomass. *Appl. Energy* 216, 662–677. <https://doi.org/10.1016/j.apenergy.2018.02.027>.
- Millinger, M., Hedenus, F., Zeyen, E., Neumann, F., Reichenberg, L., Berndes, G., 2025. Diversity of biomass usage pathways to achieve emissions targets in the European energy system. *Nat. Energy* 1–17. <https://doi.org/10.1038/s41560-024-01693-6>.
- Nunes, L.J., Silva, S., 2023. Optimization of the residual biomass supply chain: process characterization and cost analysis. *Logistics* 7 (3), 48.
- Nunes, L.J.R., Causer, T.P., Ciolkosz, D., 2020. Biomass for energy: a review on supply chain management models. *Renew. Sustain. Energy Rev.* 120, 109658. <https://doi.org/10.1016/j.rser.2019.109658>.
- Page-Dumroese, D.S., Franco, C.R., Archuleta, J.G., Taylor, M.E., Kidwell, K., High, J.C., Adam, K., 2022. Forest biomass policies and regulations in the United States of America. *Forests* 13 (9), 1415. <https://doi.org/10.3390/f13091415>.
- Parajuli, M., Gallagher, T., Cristan, R., Daniel, M.J., Mitchell, D., McDonald, T., 2023. Opportunities and challenges of woody biomass harvesting practices in the Southeastern region of the United States. *Int. J. For. Eng.* 34 (3), 303–316. <https://doi.org/10.1080/14942119.2023.2238149>.
- Parsakhoo, A., Ezzati, S., Pirov, S., & Akerdi, Y. R. (2022). Operational Planning Of Optimal Wood Supply-Chain Network with Focus on Inbound Logistics Constraints. doi: 10.21203/rs.3.rs-1444409/v1.
- Pfohl, H.C., Buse, H.P., 2000. Inter-organizational logistics systems in flexible production networks: an organizational capabilities perspective. *Int. J. Phys. Distrib. Logist. Manag.* 30 (5), 388–408. <https://doi.org/10.1108/09600030010336153>.
- Prajogo, D., Oke, A., Olhager, J., 2016. Supply chain processes: linking supply logistics integration, supply performance, lean processes and competitive performance. *Int. J. Oper. Prod. Manag.* 36 (2), 220–238. <https://doi.org/10.1108/IJOPM-03-2014-0129>.
- Rodias, E., Berruto, R., Bochtis, D., Sopegno, A., Busato, P., 2019. Green, yellow, and woody biomass supply-chain management: a review. *Energies* 12 (15), 3020. <https://doi.org/10.3390/en12153020>.
- Ramos, S.J., 2022. Biomass logistics: mythistory and sociotechnical imaginary in trans-Atlantic wood pellet assemblage. *Environ. Plann. E: Nat. Space* 5 (1), 318–339. <https://doi.org/10.1177/2514848620979311>.

- Ransikarbum, K., Pitakaso, R., 2024. Multi-objective optimization design of sustainable biofuel network with integrated fuzzy analytic hierarchy process. *Expert Syst. Appl.* 240, 122586. <https://doi.org/10.1016/j.eswa.2023.122586>.
- Ranta, T., Föhr, J., Karttunen, K., Knutas, A., 2014. Radio frequency identification and composite container technology demonstration for transporting logistics of wood biomass. *J. Renewable Sustainable Energy* 6 (1), 012301. <https://doi.org/10.1063/1.4862786>.
- Raosoftware, 2025. Sample size calculator. <http://www.raosoftware.com/samplesize.html>.
- Ravula, P.P., Grisso, R.D., Cundiff, J.S., 2008. Comparison between two policy strategies for scheduling trucks in a biomass logistic system. *Bioresour. Technol.* 99 (13), 5710–5721. <https://doi.org/10.1016/j.biortech.2007.10.044>.
- Rentizelas, A.A., Tolis, A.J., Tatsiopoulos, I.P., 2009. Logistics issues of biomass: the storage problem and the multi-biomass supply chain. *Renew. Sustain. Energy Rev.* 13 (4), 887–894. <https://doi.org/10.1016/j.rser.2008.01.003>.
- Romano, P., 2003. Co-ordination and integration mechanisms to manage logistics processes across supply networks. *J. Purch. Supply Manag.* 9 (3), 119–134. [https://doi.org/10.1016/S1478-4092\(03\)00008-6](https://doi.org/10.1016/S1478-4092(03)00008-6).
- Ruiz, J.A., Juárez, M.C., Morales, M.P., Muñoz, P., Mendiñvil, M.A., 2013. Biomass logistics: financial & environmental costs. Case study: 2 MW electrical power plants. *Biomass Bioenergy* 56, 260–267. <https://doi.org/10.1016/j.biombioe.2013.05.014>.
- Searcy, E., Hess, J.R., Tumuluru, J., Ovard, L., Muth, D.J., Trømborg, E., ... Hoefnagels, R., 2014. Optimization of biomass transport and logistics. In *International Bioenergy Trade: History, Status & Outlook on Securing Sustainable Bioenergy Supply, Demand and Markets* (pp. 103–123). doi: 10.1007/978-94-007-6982-3_5.
- Shabani, N., Sowlati, T., 2013. A mixed integer non-linear programming model for tactical value chain optimization of a wood biomass power plant. *Appl. Energy* 104, 353–361. <https://doi.org/10.1016/j.apenergy.2012.11.013>.
- Sokhansanj, S., Kumar, A., Turhollow, A.F., 2006. Development and implementation of integrated biomass supply analysis and logistics model (IBSAL). *Biomass Bioenergy* 30 (10), 838–847. <https://doi.org/10.1016/j.biombioe.2006.04.004>.
- Sokhansanj, S., Mani, S., Tagore, S., Turhollow, A.F., 2010. Techno-economic analysis of using corn stover to supply heat and power to a corn ethanol plant—part 1: cost of feedstock supply logistics. *Biomass Bioenergy* 34 (1), 75–81. <https://doi.org/10.1016/j.biombioe.2009.10.011>.
- Sokhansanj, S., Mani, S., Turhollow, A., Kumar, A., Bransby, D., Lynd, L., Laser, M., 2009. Large-scale production, harvest and logistics of switchgrass (*Panicum virgatum* L.)—current technology and envisioning a mature technology. *Biofuels Bioprod. Biorefin.* 3 (2), 124–141. <https://doi.org/10.1002/bbb.129>.
- Song, N., Aguilar, F.X., 2017. Woody biomass energy efficiency pathways: Public policy implications. *Biofuels* 8 (4), 473–483. <https://doi.org/10.1080/17597269.2016.1259520>.
- Sosa, A., Acuna, M., McDonnell, K., Devlin, G., 2015. Controlling moisture content and truck configurations to model and optimize biomass supply chain logistics in Ireland. *Appl. Energy* 137, 338–351. <https://doi.org/10.1016/j.apenergy.2014.10.018>.
- Stephen, J.D., Mabee, W.E., Saddler, J.N., 2010. Biomass logistics as a determinant of second-generation biofuel facility scale, location and technology selection. *Biofuels Bioprod. Biorefin.* 4 (5), 503–518. <https://doi.org/10.1002/bbb.239>.
- Svanberg, M., Olofsson, I., Flodén, J., Nordin, A., 2013. Analysing biomass torrefaction supply chain costs. *Bioresour. Technol.* 142, 287–296. <https://doi.org/10.1016/j.biortech.2013.05.048>.
- Takki, K., 2024. Biomass to Account for 18% of the Energy Market, <https://statzon.com/insights/the-global-biomass-market>.
- Tégla, Z., Hágen, I., Holló, E., Takácsné, G.K., 2012. Adoption of logistic principles in woody-biomass energy clusters. *Rev. Appl. Socio-Econ. Res., REASER* 4 (2), 236–246.
- Tracey, M., Lim, J.S., Vonderembse, M.A., 2005. The impact of supply-chain management capabilities on business performance. *Supply Chain Manage. Int. J.* 10 (3), 179–191. <https://doi.org/10.1108/13598540510606232>.
- Verma, R., 2024. Supply chain management: optimizing efficiency and sustainability in the supply network. *J. Adv. Manage. Stud.* 1 (2), 13–18.
- WBA, 2018. Biomass supply chains. Harvesting & collection, pre-treatment and upgrading, storage, transportation & handling, World Bioenergy Association, <https://www.worldbioenergy.org/uploads/Factsheet%20-%20Biomass%20Supply%20Chains.pdf>.
- Wohlin, C., 2014. May). guidelines for snowballing in systematic literature studies and a replication in software engineering. In: In Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering, pp. 1–10. <https://doi.org/10.1145/2601248.260126>.
- Xie, F., Huang, Y., Eksioğlu, S., 2014. Integrating multimodal transport into cellulosic biofuel supply chain design under feedstock seasonality with a case study based on California. *Bioresour. Technol.* 152, 15–23. <https://doi.org/10.1016/j.biortech.2013.10.074>.
- Zamora-Cristales, R., Sessions, J., Boston, K., Murphy, G., 2015. Economic optimization of forest biomass processing and transport in the Pacific Northwest USA. *Forest Science* 61 (2), 220–234.
- Zhu, X., Li, X., Yao, Q., Chen, Y., 2011. Challenges and models in supporting logistics system design for dedicated-biomass-based bioenergy industry. *Bioresour. Technol.* 102 (2), 1344–1351. <https://doi.org/10.1016/j.biortech.2011>.
- Han, H., Chung, W., & Wells, L. (2015). A mathematical approach to biomass feedstock logistics problems. In *38th Annual COFE Meeting* (pp. 19–22).